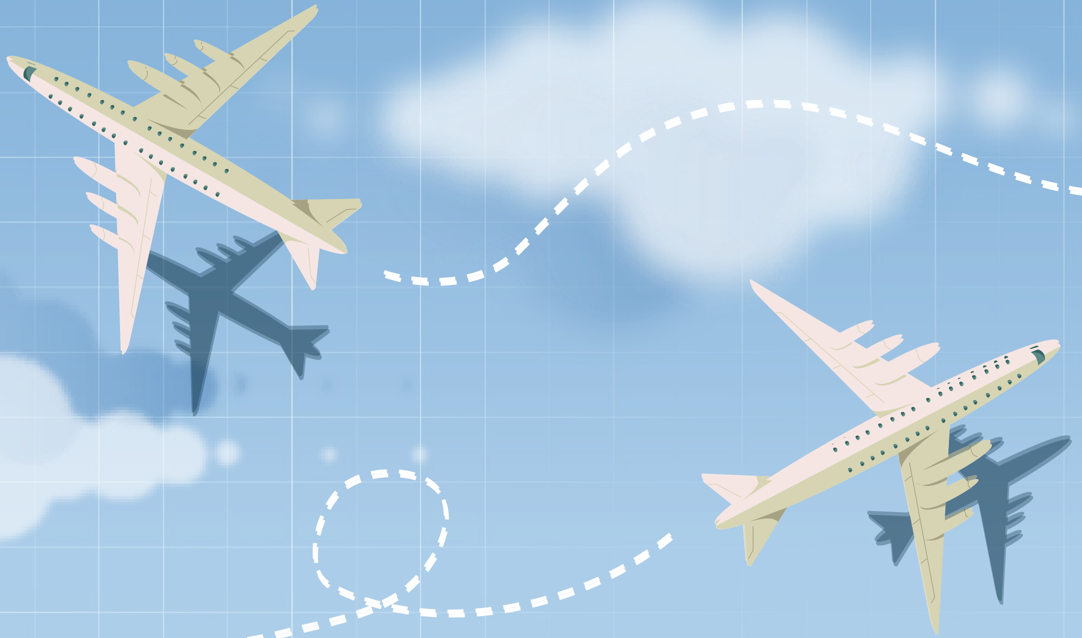
Biomass-to-Jet: Using Gasification and Fischer–Tropsch Chemistry

Anna Cai, Qussay Suleiman, Miguel Piza, Jazmin Aca



01 introduction

Project overview

01

Woody biomass

02

Syngas production

Biomass gasification to $CO + H_2$

03

Syngas cleaning

Water–gas shift, cooling, CO_2 removal

04

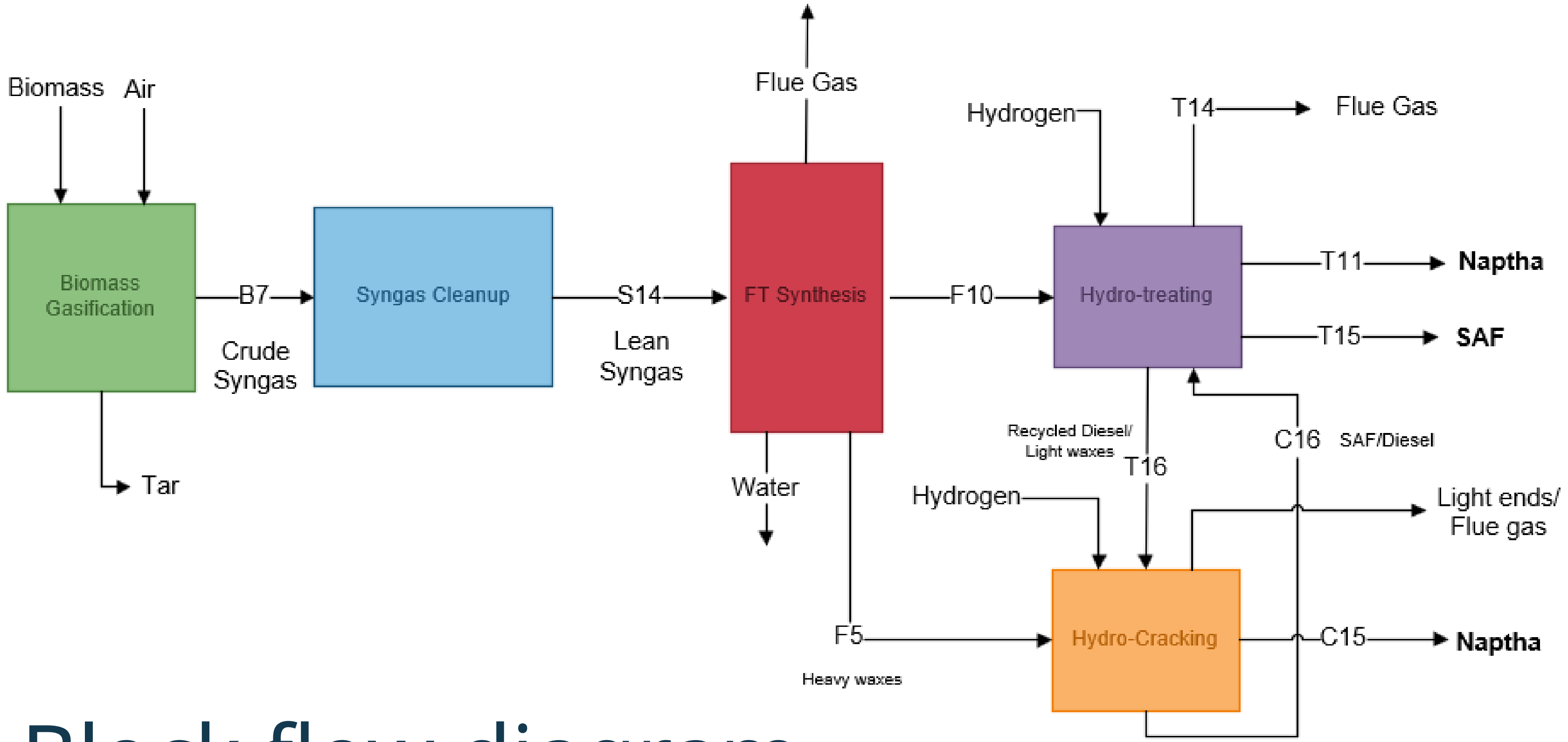
Fischer-tropsch synthesis

Syngas → FT waxes + middle distillates

05

Upgrading and fractionation

Hydrocracking & Hydrotreating

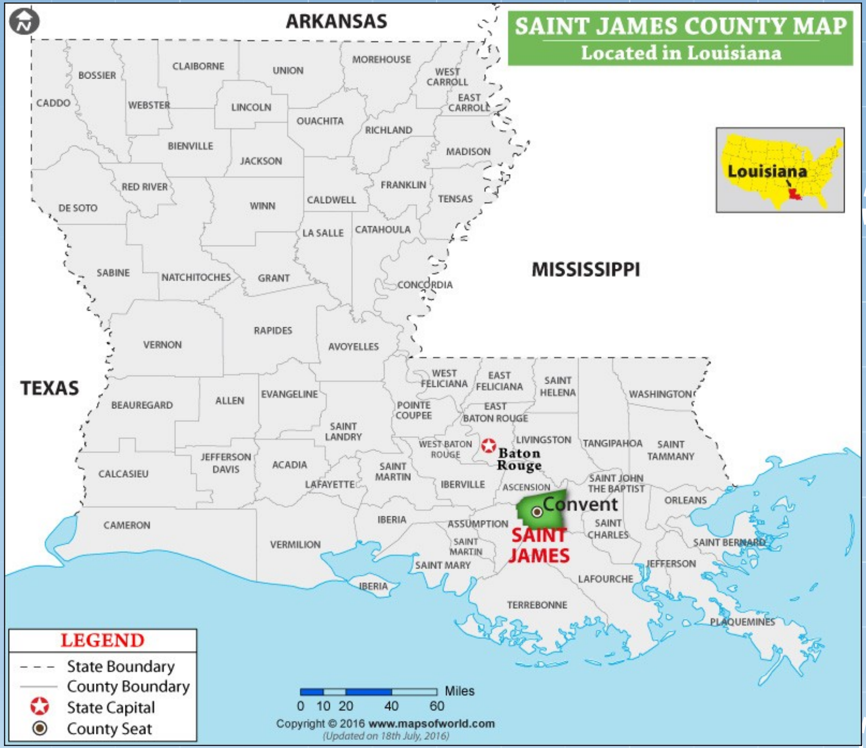


Block flow diagram

Design basis

Plant Capacity
- 10,000 barrels per day of SAF
- 3,500 barrels per day of Naphtha

Location: Saint James County, Louisiana (U.S. Gulf Coast)
- Abundant forestry and agricultural biomass
- Strong hydrogen availability
- Deep-water port access (New Orleans, Baton Rouge)
- High SAF demand from nearby major airports



Malik, V. *St James Parish Map, Louisiana*

Project updates

01

Tar reforming

Added Tar to the gasifier and tar reformer

02

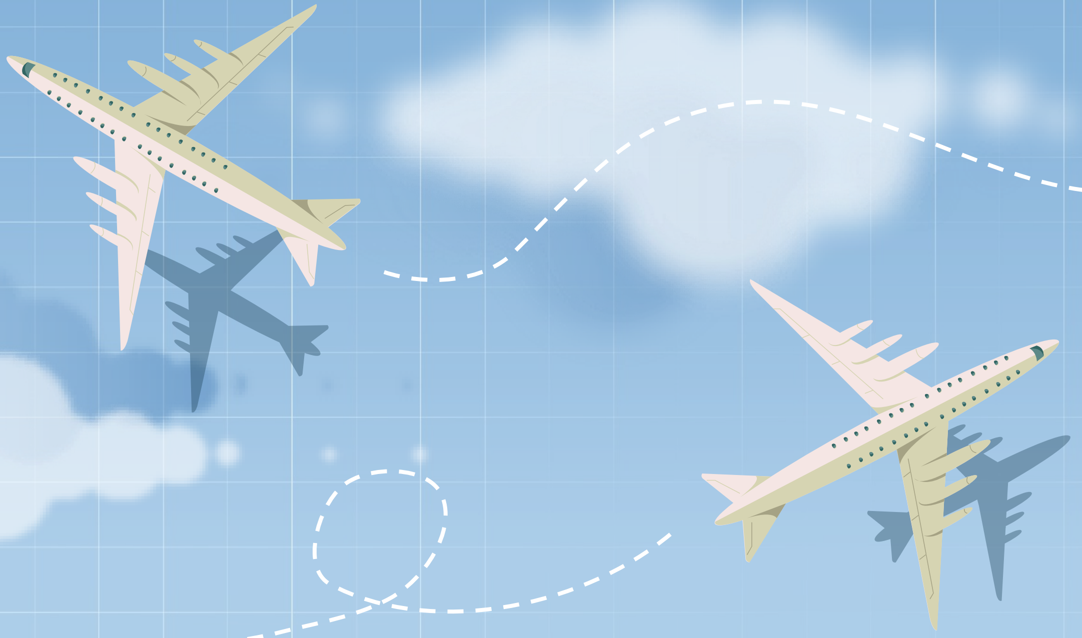
Steam Generatio n

Incorporated steam generation throughout the process

03

Cooling water

Greatly reduced cooling water usage



02

safety

Safety and environmental analysis

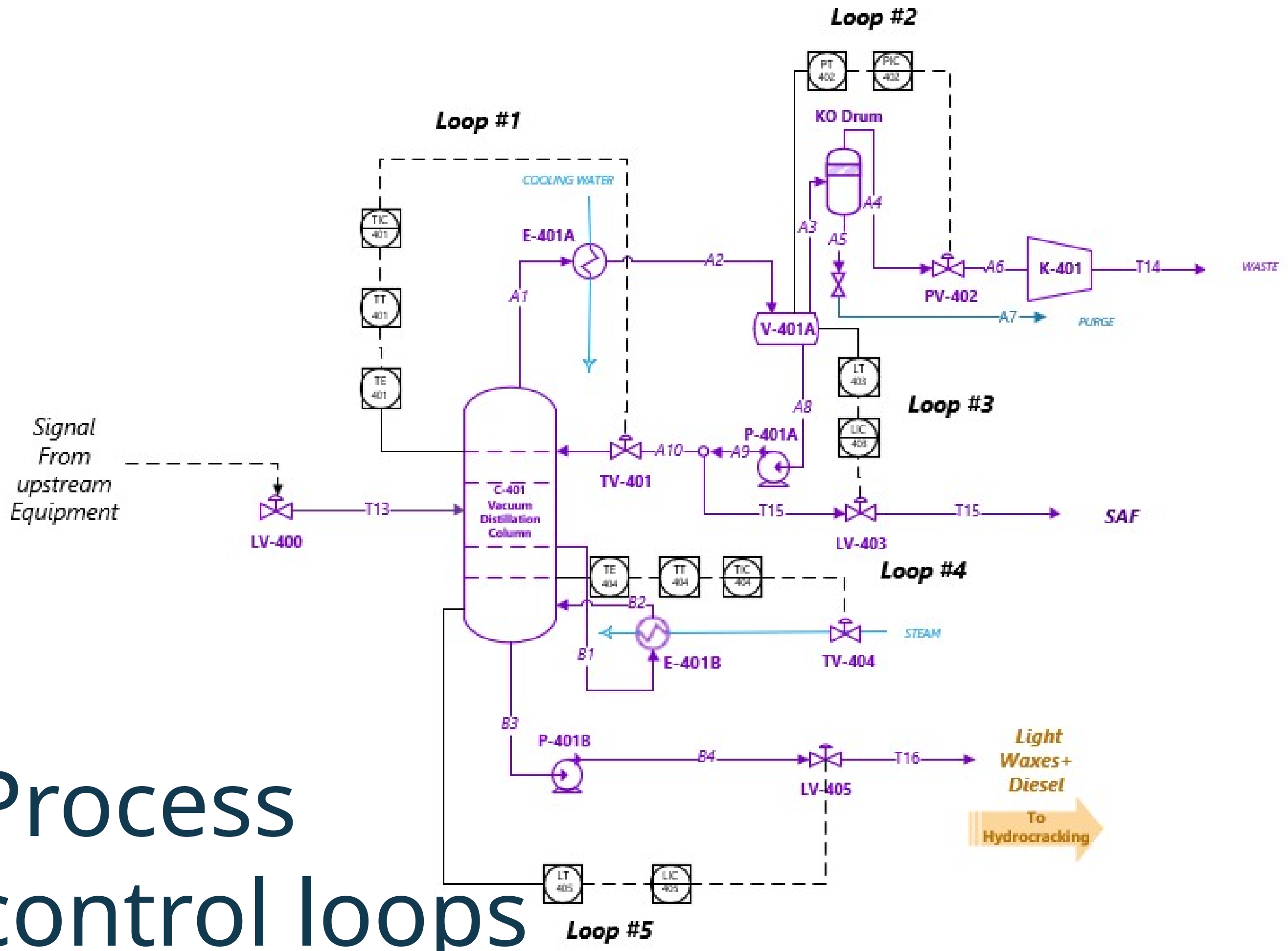
Component	Safety/Exposure Limits
Hydrogen	1. Highly flammable gas (Level 4) 2. Oxygen displacement hazard
Carbon Monoxide	1. Toxic if inhaled 2. Exposure limits TWA: 50 ppm over 8 hours
Carbon Dioxide	1. Respiratory Hazard 2. Exposure limit: 5000 ppm
Oxygen	1. Strong Oxidizer 2. Increases combustion and fire severity
Jet Fuel	1. Highly flammable liquid 2. Exposure limit TWA: 100 mg/m^3 over 8 hours
Naphtha	1. Highly flammable 2. Exposure limit TWA: 300 ppm over 8 hours

Operational safety concerns

Unit	Safety Concerns	Prevention and Mitigation
Biomass Dryer	Fire or Explosion Conditions	Temperature monitoring and high temperature shutdown.
	Dust from woodchips	Install dust suppression, proper ventilation, and enforce proper respiration protection
Biomass Gasifier	High Temperatures	Temperature monitoring and high temperature shutdown.
	Flammable gasses	Gas leak detection
Water gas shift reactor	Temperature rise	Tight temperature monitoring across bed
	Overpressure	Pressure relief valves
	CO Leaks	Use CO detection around reactor area
CO_2 Absorber	CO_2-rich solvent leaks	Provide secondary containment for possible solvent leaks
	Column Flooding	Differential Pressure monitoring
Fischer-Tropsch Reactor	Highly Exothermic Reaction	Temperature measurements across reactor
	CO and H_2 Leaks	Feed control, pressure relief to safe disposal system
	Runaway temperatures	High-high temperature emergency shutdown
Hydrotreating Reactor	High pressure hydrogen and hydrocarbon mixture	High-high temperature and pressure trips

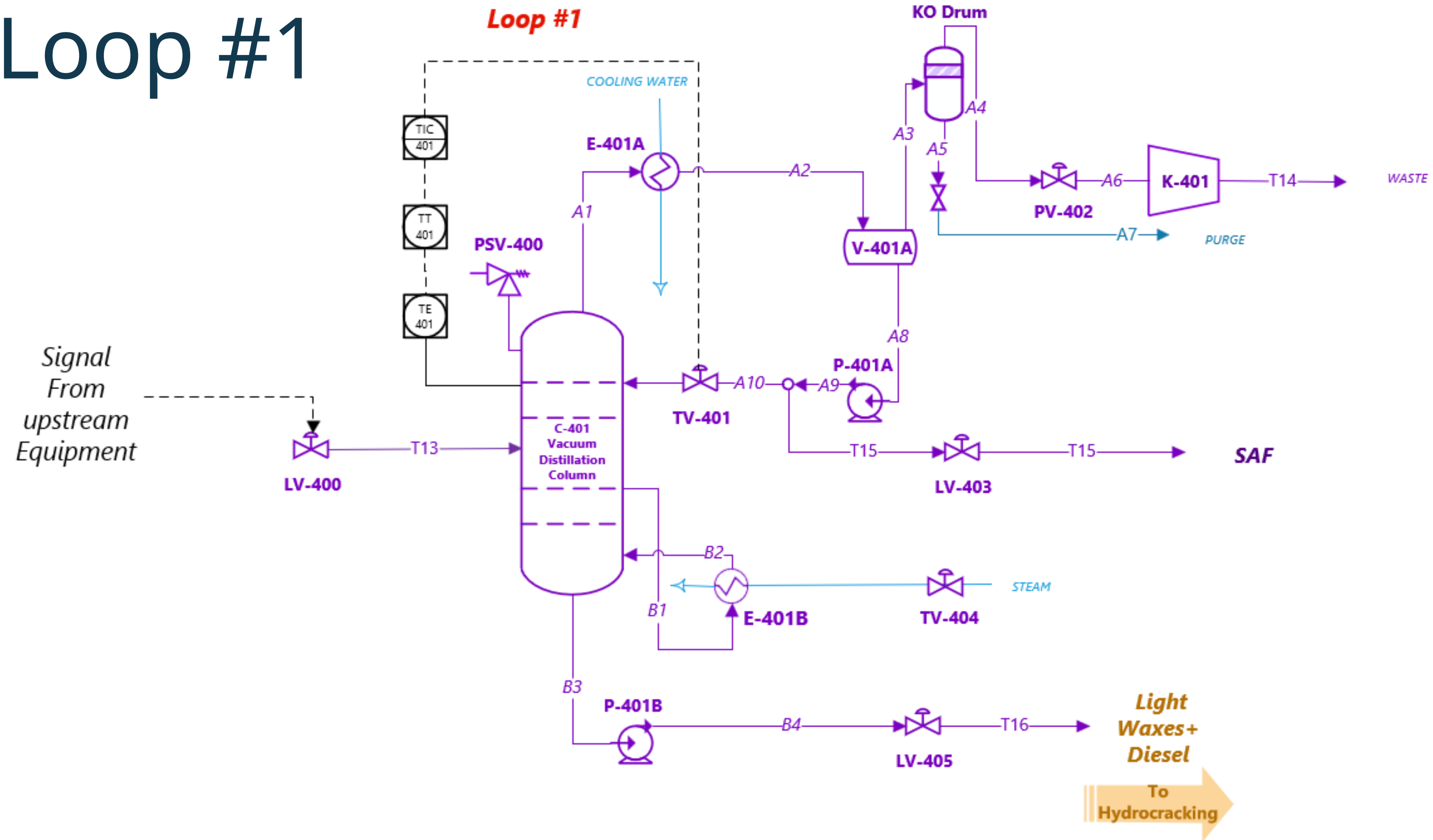
Operational safety concerns

Unit	Safety Concerns	Prevention and Mitigation
Hydrotreater Distillation Column	Flammable hydrocarbon inventory	Tight pressure, temperature and level control
	Hot Product and Bottoms release	Insulate hot surfaces
	Flooding	DP monitoring
Hydrotreater Vacuum Distillation Column	Vacuum Loss	Vacuum monitoring and alarms
Hydrocracker	High pressure hydrogen	Route relief streams safely
	Strong exothermic reaction potential hotspot formation	Multiple temperature measurements
	Hydrogen leak and fire risk	Leak detection and monitoring, flammable gas detectors
3-phase separator	Hydrogen rich gas release	Closed vent handling
Hydrotreater Distillation Column	Flammable hydrocarbon inventory	Tight pressure, temperature and level control
	Hot Product and Bottoms release	Insulate hot surfaces
	Over head vapor release	Safe overhead vent handling

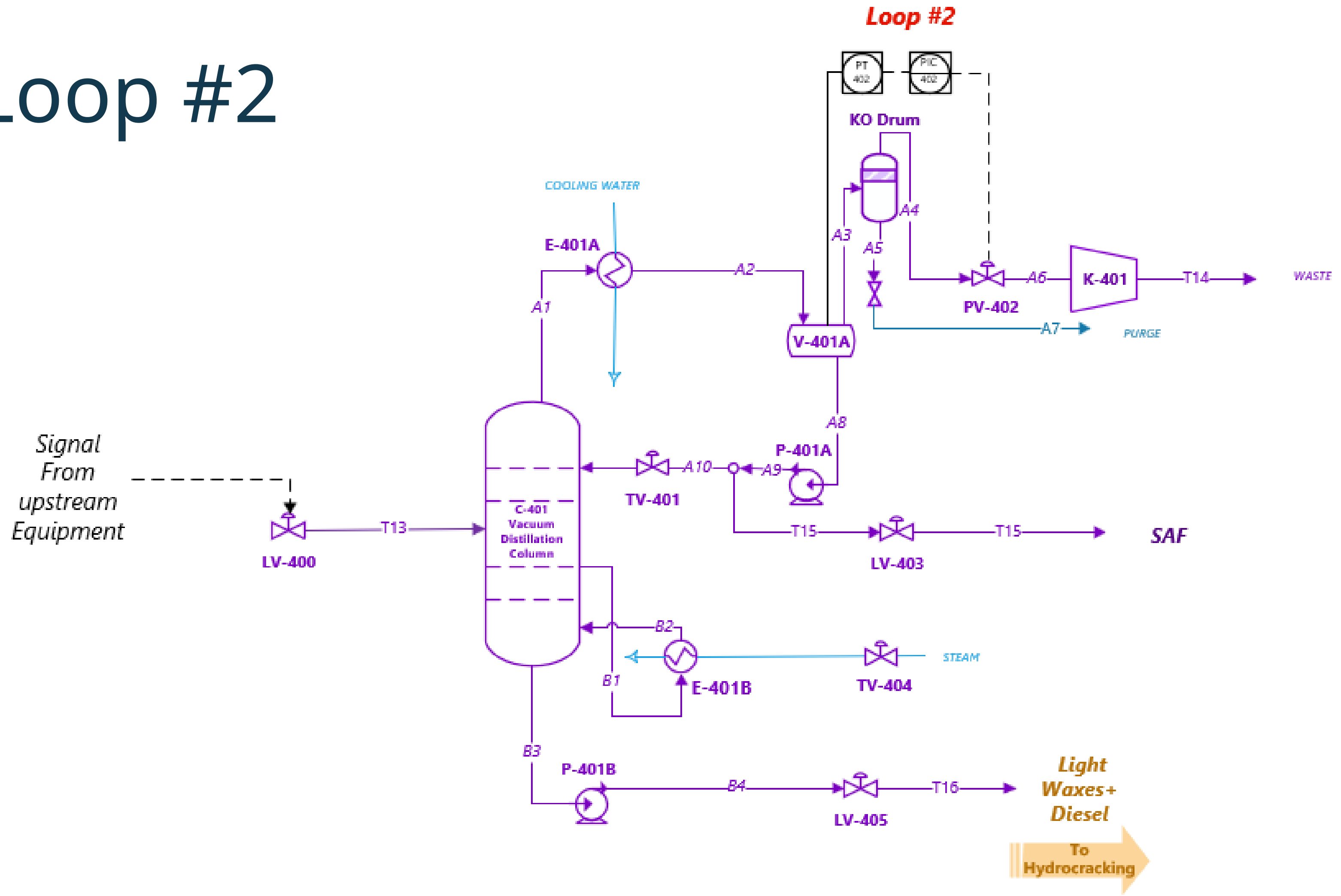


Process control loops

Loop #1

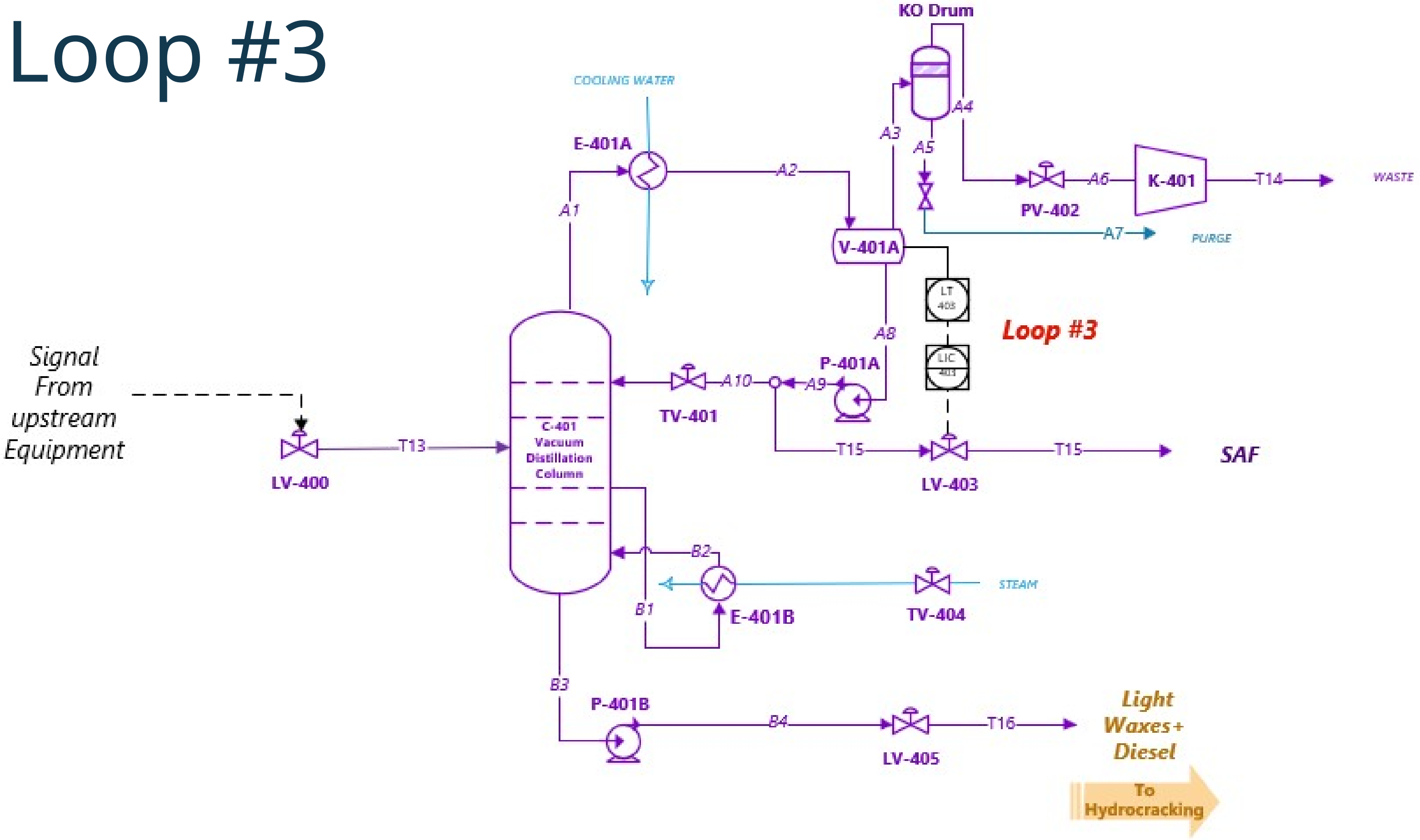

Loop #1
KO Drum
COOLING WATER
TIC 401
TT 401
TE 401
E-401A
PSV-400
A1
A2
A3
A4
A5
A6
A7
A8
A9
A10
V-401A
PV-402
K-401
T14
WASTE
PURGE
P-401A
TV-401
Signal From upstream Equipment
LV-400
T13
C-401 Vacuum Distillation Column
T15
LV-403
SAF
B1
B2
B3
B4
E-401B
TV-404
STEAM
P-401B
LV-405
T16
Light Waxes+ Diesel
To Hydrocracking

Loop #2

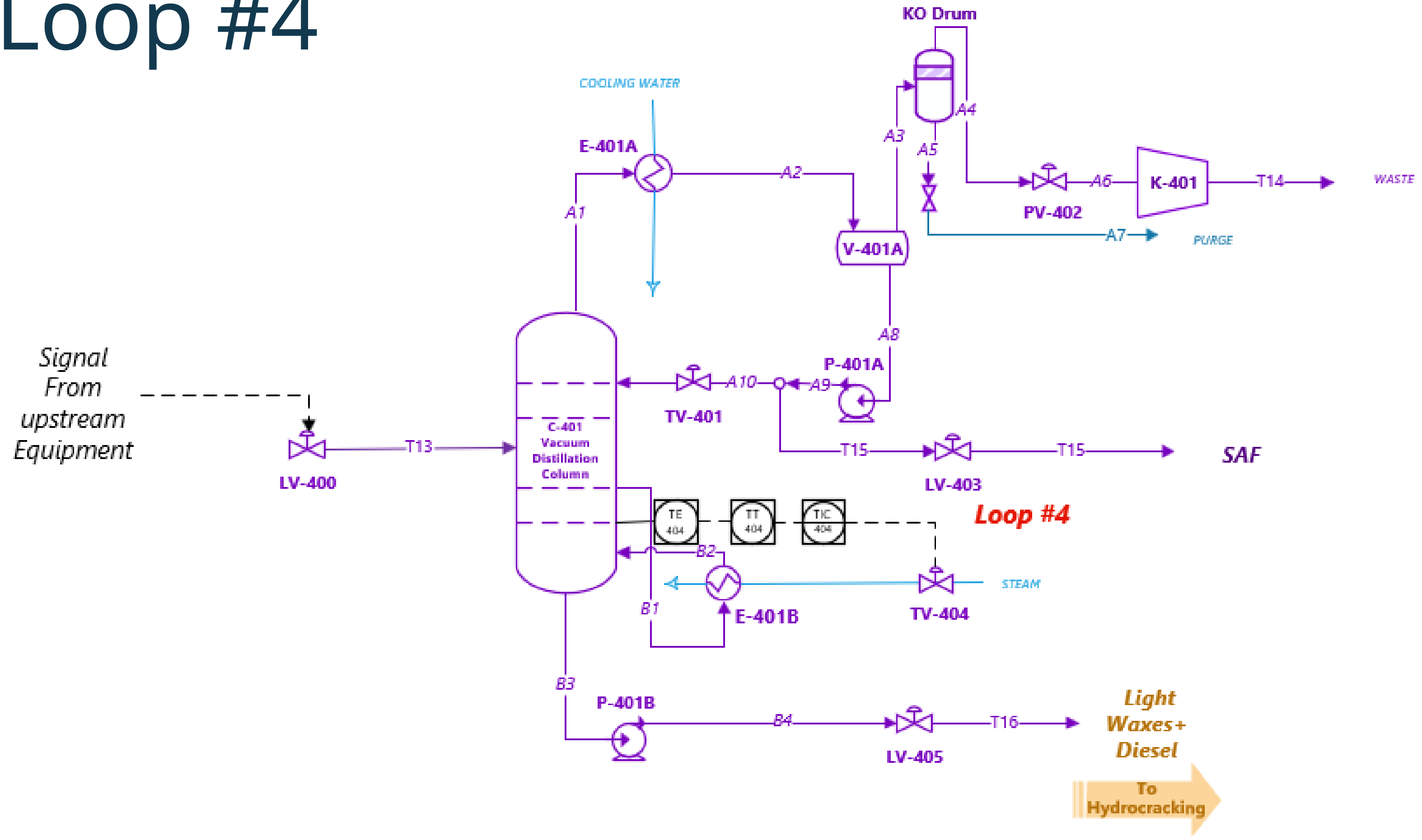


Loop #2
PT 402
PIC 402
KO Drum
COOLING WATER
E-401A
A1
A2
A3
A4
A5
A6
A7
A8
A9
A10
V-401A
PV-402
K-401
T14
WASTE
PURGE
Signal From upstream Equipment
LV-400
T13
C-401 Vacuum Distillation Column
TV-401
P-401A
T15
LV-403
SAF
B1
B2
E-401B
STEAM
TV-404
B3
P-401B
B4
LV-405
T16
Light Waxes+ Diesel
To Hydrocracking

Loop #3

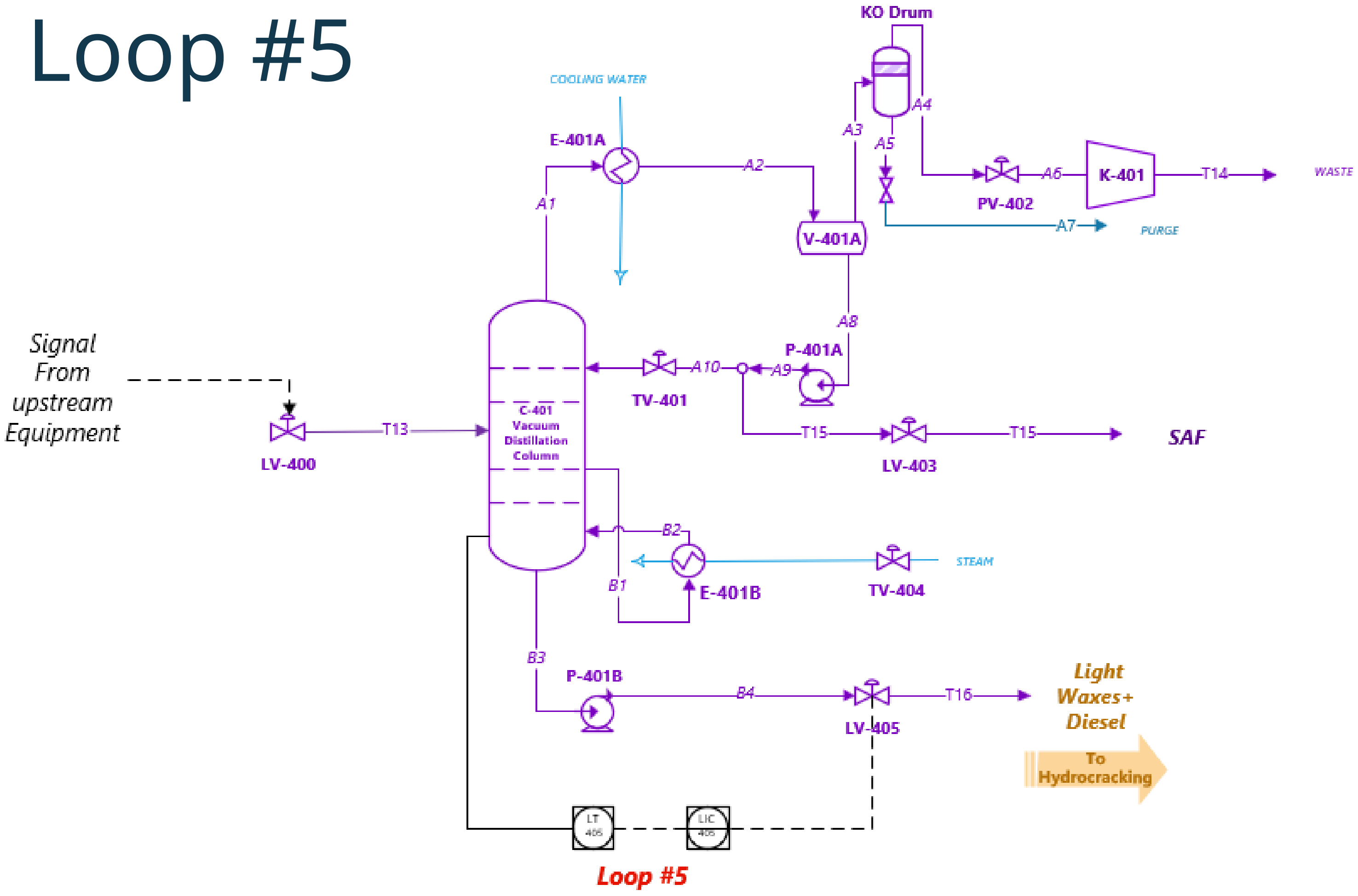

KO Drum
COOLING WATER
E-401A
A1
A2
A3
A4
A5
A6
A7
A8
A9
A10
PV-402
K-401
T14
WASTE
PURGE
V-401A
LT 403
LIC 403
Loop #3
P-401A
TV-401
Signal From upstream Equipment
LV-400
T13
C-401 Vacuum Distillation Column
T15
LV-403
SAF
B1
B2
B3
B4
E-401B
TV-404
STEAM
P-401B
LV-405
T16
Light Waxes+ Diesel
To Hydrocracking

Loop #4



KO Drum
COOLING WATER
E-401A
A1
A2
A3
A4
A5
A6
A7
A8
A9
A10
V-401A
PV-402
K-401
T14
WASTE
PURGE
Signal From upstream Equipment
LV-400
T13
C-401 Vacuum Distillation Column
P-401A
TV-401
T15
LV-403
SAF
TE 404
TT 404
TIC 404
Loop #4
B1
B2
B3
B4
E-401B
STEAM
TV-404
P-401B
LV-405
T16
Light Waxes+ Diesel
To Hydrocracking

Loop #5


KO Drum
COOLING WATER
E-401A
A1
A2
A3
A4
A5
A6
A7
A8
A9
A10
PV-402
K-401
T14
WASTE
PURGE
V-401A
P-401A
TV-401
Signal From upstream Equipment
LV-400
T13
C-401 Vacuum Distillation Column
T15
LV-403
SAF
B1
B2
E-401B
STEAM
TV-404
B3
P-401B
B4
T16
LV-405
Light Waxes+ Diesel
To Hydrocracking
LT 405
LIC 405
Loop #5

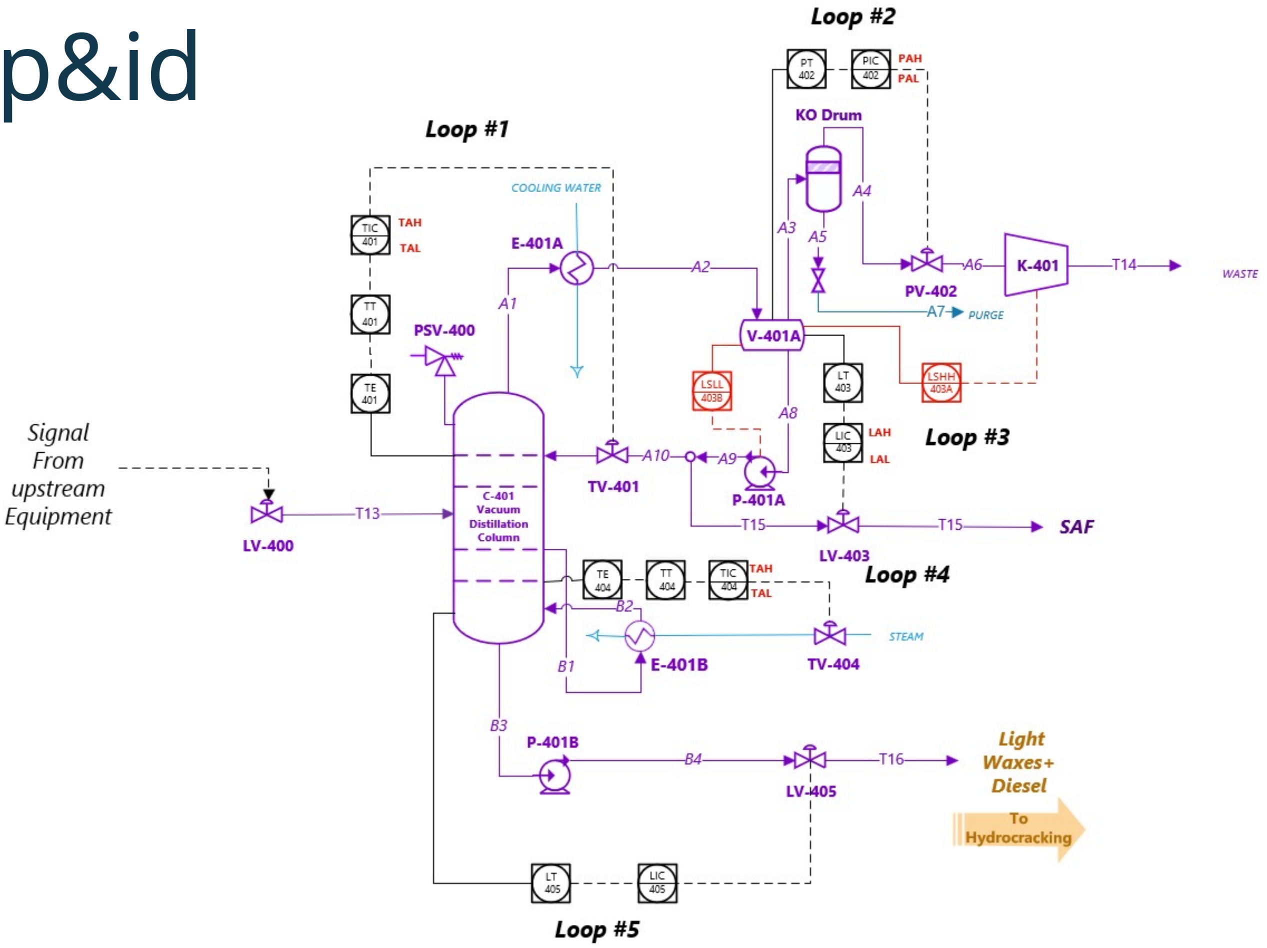
Hazard and operability study

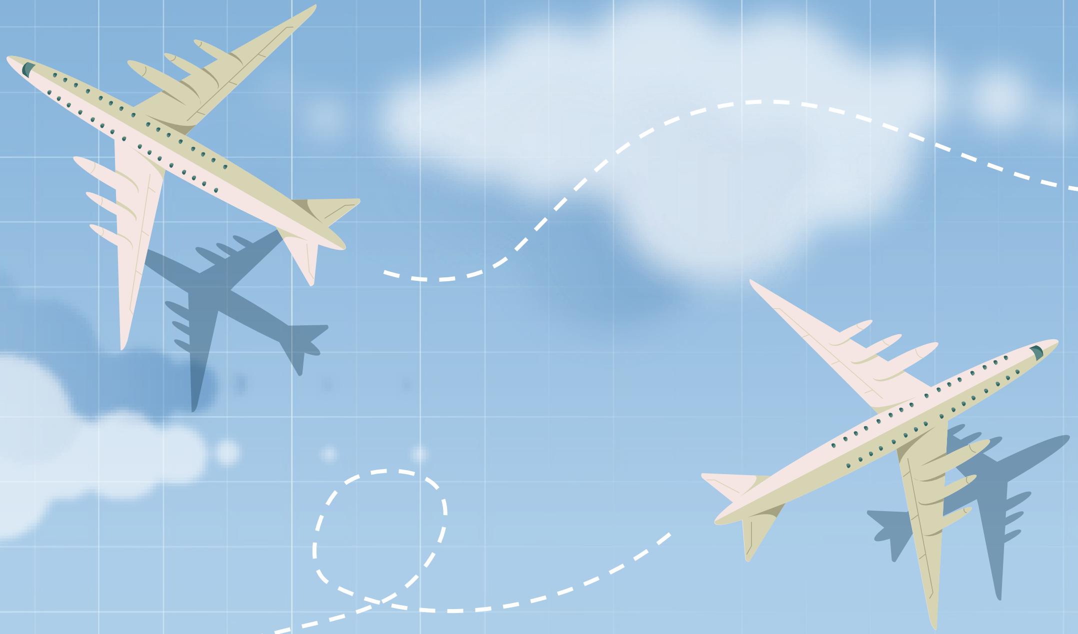
Deviation	Consequence	Cause	Safeguard	Recommended Action
High Pressure	Vessel mechanically fails due to overpressure and potential fatality.	TV-404 fails open, excess steam flow provides more heat to the reboiler than the overhead condenser system can remove. SAF components no longer condense, vacuum pump not rated for higher flow of vapor, pressure builds to MAWP, vessel failure, loss of process containment and potential fatality.	- PT-402 readout in control room. - PSV-001 safety relief valve. - TT-401 readout in control room	- Add PAH-402 - Add TAH-401
Low Pressure	Vessel fails due to high vacuum, collapse	PV-402 fails open, excess flow through K-401 vacuum pump, pressure drops below vacuum set-point, potential to fail column, potential fatality.	- Column is designed for full vacuum (FV).	- Add PAL-402
LESS Flow	No new consequences see high pressure	TV-401 fails closed, loss of reflux, excess heat to column, potential for overpressure to MAWP, vessel failure, loss of process containment and potential fatality.	- PT-402 readout in control room. - PSV-001 safety relief valve. - TT-401 readout in control room	- Add PAH-402 - Add TAH-401
MORE Flow	No new consequences, see high pressure	TV-404 fails open, causing excessive vapor traffic through the column	- TT-401 readout in the control - -PSV-001 safety relief valve	- Add PAH-402 - Add-TAH-401

Hazard and operability study

Deviation	Consequence	Cause	Safeguard	Recommended Action
Low Level	See high pressure for ultimate consequence. Evaluate seal fire here.	LV-403 fails open, excess material pumped from V-401A, loss of level, pump P-401A runs dry, equipment damage, potential seal fire, potential injury. Loss of reflux also occurs, leading to high pressure.	- TT-401 readout in control room	- Add TAH-401 - Install redundant level devices with alarms. - Install double mechanical seal with seal routed to flare, high pressure alarms on seal pot.
High Level	No new consequences, see low pressure	LV-403 fails closed, material accumulates in V-401A and be enrtrained towards K-401, potential damage to P-401A, loss of vacuum pressure	- PT-402 readout in the control room - KO drum	- Add LAH-403 - Consider adding LAHH to isolate K-401
Low Temperature	No new consequences, see low pressure	TV-404 fails closed, TIC-404 reads low, or TT-404 reads high. TV-401 fails open	- PT-402 readout in the control room - TT-401	- Add TAL-401
High Temperature	No new consequences, see high pressure	TV-404 fails open, TIC-404 can read high, or TT-404 can read low, allowing excess steam to E-401B TV-401 fails closed, loss of reflux, excess heat to column, potential for overpressure to MAWP, vessel failure, loss of containment and potential fatality	- PT-402 readout in the control room - TT-401 readout in control room - PSV-400 pressure relief valve	- Add TAH-404 - Consider TAHH to isolate steam - Consider adding LAL to cooling water flow on E-401A

p&id

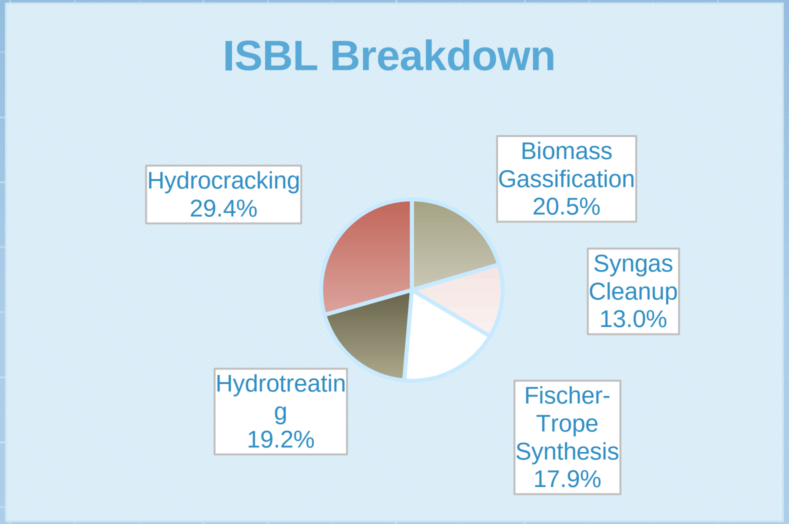

Loop #1
Loop #2
Loop #3
Loop #4
Loop #5
TIC 401
TAH
TAL
TT 401
TE 401
COOLING WATER
E-401A
A1
A2
PSV-400
Signal From upstream Equipment
LV-400
T13
C-401 Vacuum Distillation Column
TV-401
A10
A9
P-401A
PT 402
PIC 402
PAH
PAL
KO Drum
A3
A4
A5
PV-402
A6
K-401
T14
WASTE
A7
PURGE
V-401A
LSLL 403B
LT 403
LSHH 403A
A8
LIC 403
LAH
LAL
T15
LV-403
SAF
TE 404
TT 404
TIC 404
B2
B1
E-401B
TV-404
STEAM
B3
P-401B
B4
LV-405
T16
Light Waxes+ Diesel
To Hydrocracking
LT 405
LIC 405

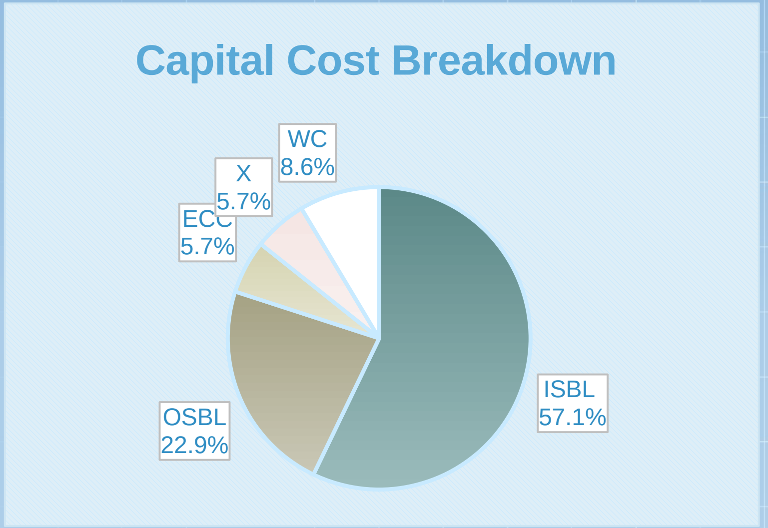


03

Process economics

Capital costs





Type	ISBL	OSBL	ECC	X	WC	TCC
Cost ($MM)	112.1	44.8	11.2	11.2	16.8	196.1

Variable Cost of production

Raw Material	Amount (lb/hr)	Cost rate ($/lb)	Cost ($MM/yr)
Biomass	524,000	0.03	125.6
Green Hydrogen	3500	3.4	95.5
Oxygen	11,000	0.05	4.4
Total	**225.5 $MM/oper-yr**		

Equipment	Catalyst Type	Cost ($/yr)
FT reactor	LT Iron	90,000
Hydrotreater	$CoMo/Al_2O_3$	45,000
Hydrocracker	$NiMo/Al_2O_3$ and NiW/Zeolite	170,000
Total	**305,000 $/oper-yr**	

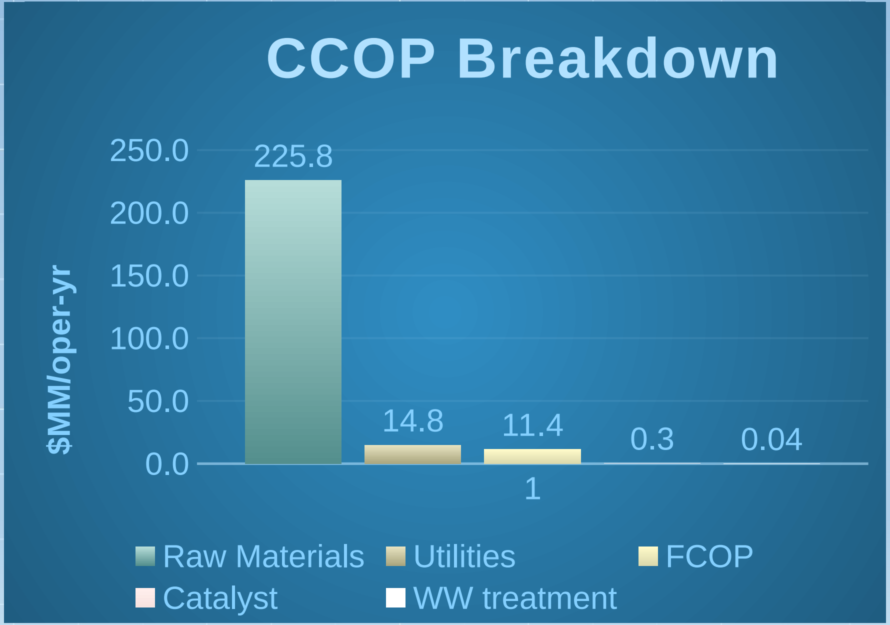
Utility	Amount	Cost rate	Cost ($MM/yr)
Cooling Water	15,900 gpm	0.0354 $/GJ	0.43
Natural Gas	177,000 lb/hr	6.0 $/GJ	6.2
Electricity	18,350 kWh	0.0561 $/kWh	8.2
WW treatment	2500 gal/hr	2 $/1000 gal	0.04
Total	**14.8 $MM/oper-yr**		

VCOP	**240.7 $MM/yr**

Fixed cost of production

Number of Shifts per day	3
Number of Operators per shift	5
Operator Salary ($/yr)	$70,000
Total Operator Compensation ($/yr)	1.05 $MM
Supervision (25% of Operating Labor), ($/yr)	0.26 $MM
Direct Salary Overhead (45% of Labor and Superv.), ($/yr)	0.59 $MM
Maintenance (3% ISBL), ($/yr)	3.36 $MM
Overhead Expenses (Plant, Taxes, Insurance, land), ($/yr)	6.11 $MM
Total FCOP	**11.38 $MM**

Cash Cost of production



Type	Cost ($MM/yr)
Raw Materials	225.8
Utilities	14.8
Catalyst	0.3
WW treatment	0.04
FCOP	11.4
CCOP	**252.1**

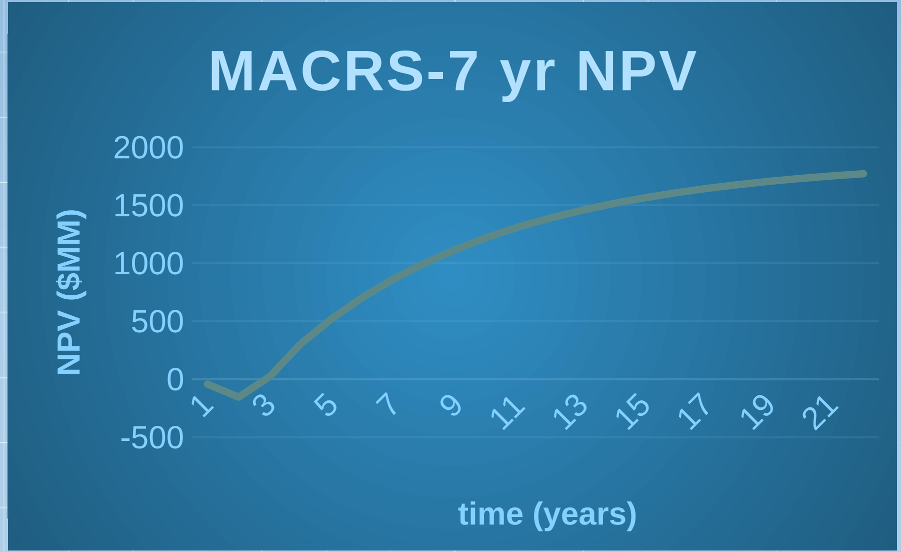
Revenue and gross profit

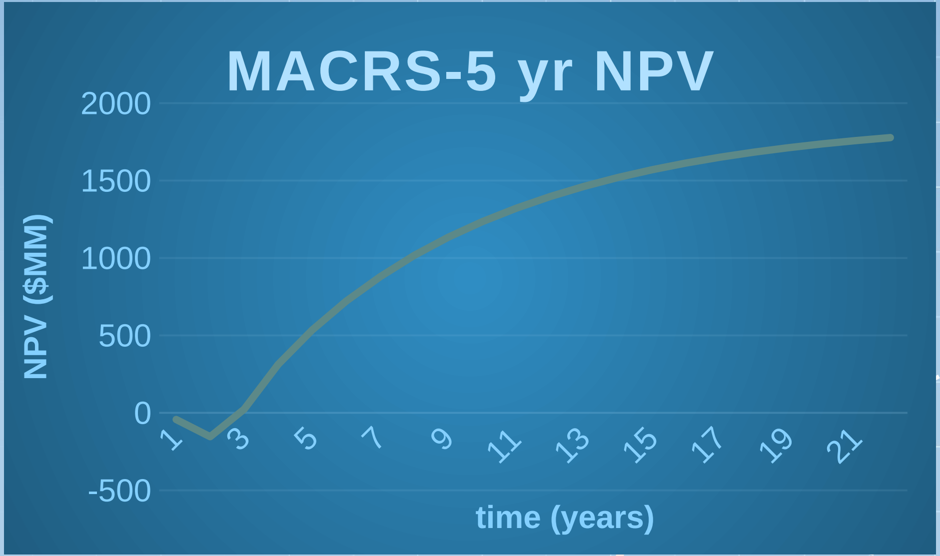
Product Type	Revenue ($MM/yr)
SAF	438.5
Naphtha	135.0
Total	**573.5**

Credit Type	Revenue ($MM/yr)
SAF	198.0
Naphtha	31.8
Total	**229.8**

Total Revenue	**803.3 $MM/yr**
Gross Profit	**551.2 $MM/yr**

Dcfror and npv

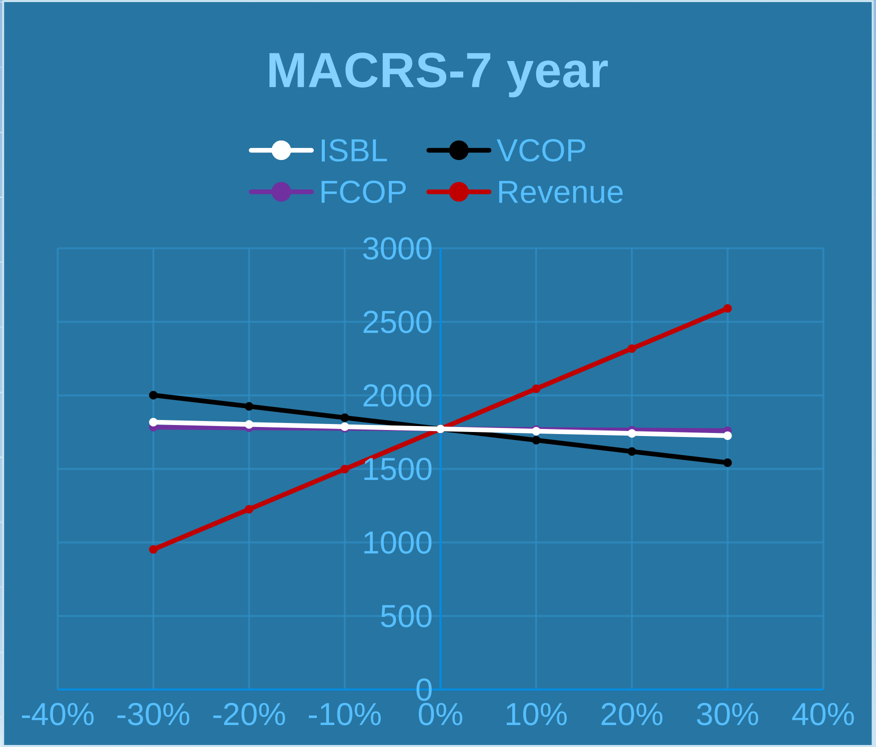


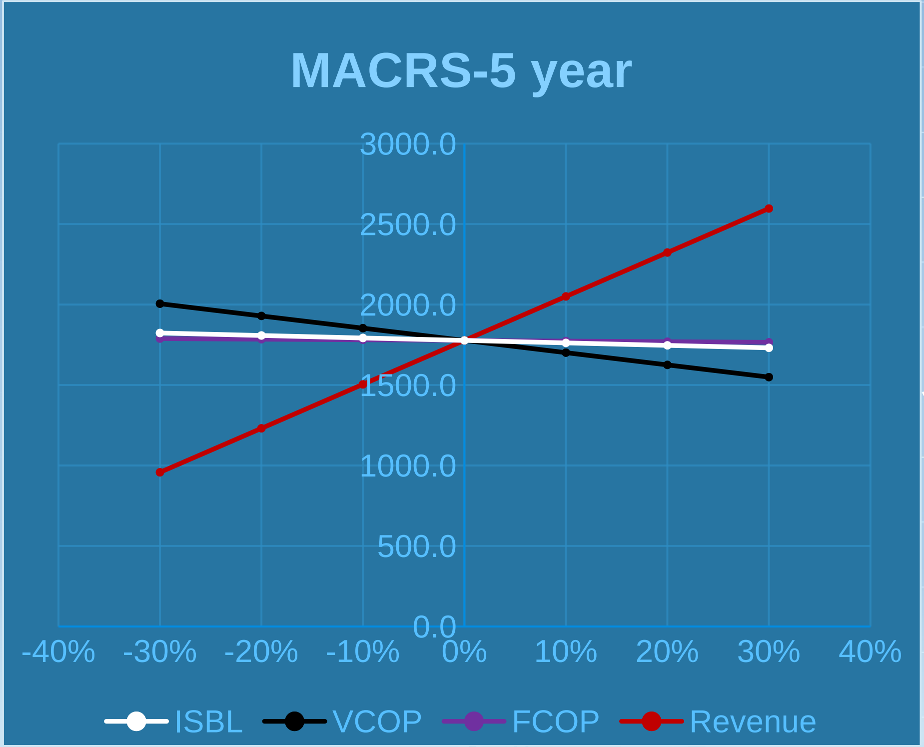


Constants	
Tax Rate	26.5%
Interest Rate	15%

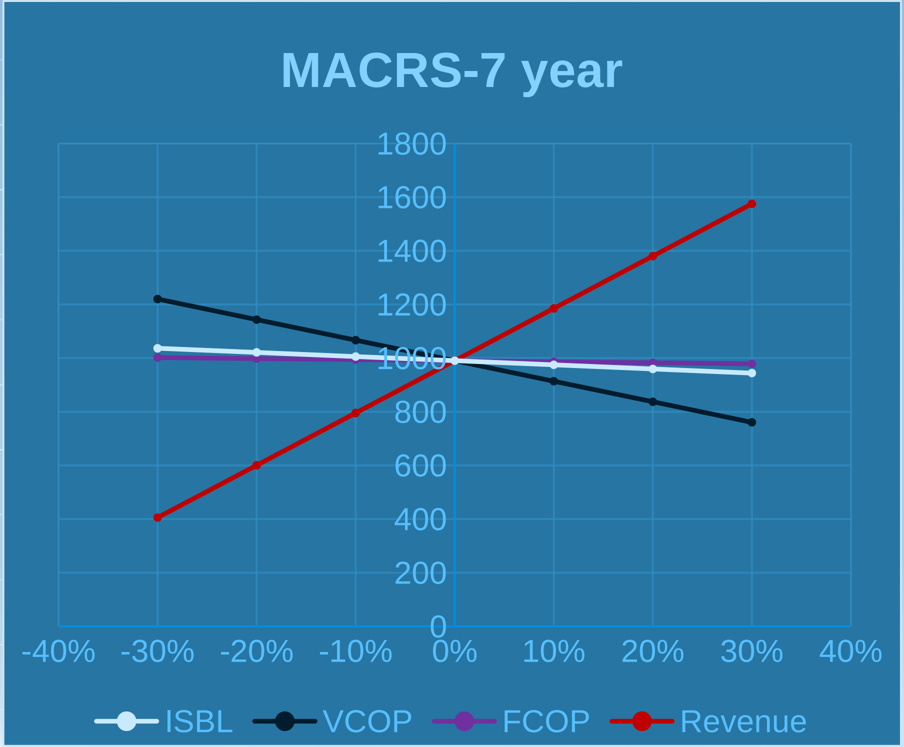
DCFOR (7-yr)	**135.3%**
DCFOR (5-yr)	**136.1%**

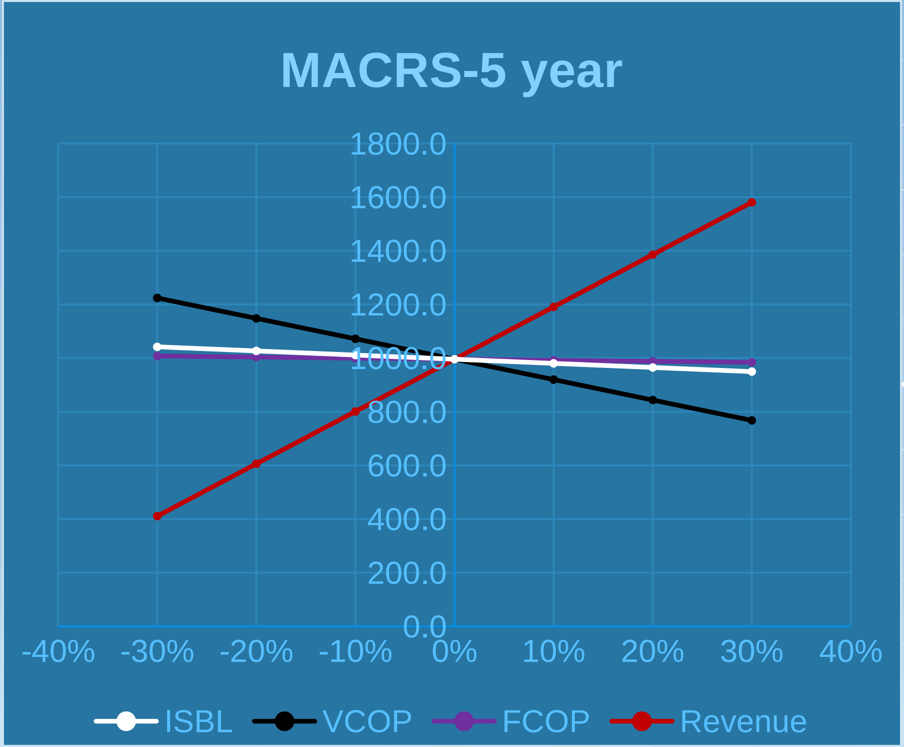
Sensitivity





Sensitivity analysis (No tax





Thank You!

Any Questions?

References

1. **Béalu, Z.; Burger, J.; Friedemann, M.** *Process Simulation and Optimization of Alternative Liquid Fuels Production: A Techno-Economic Assessment of the Production of HEFA Jet Fuel*; Master's Thesis, 2017. Available online: https://elib.dlr.de/117806/1/Masterarbeit_Zoe_Bealu.pdf (accessed 2025).

2. **Curcio, E.** Techno-Economic Analysis of Hydrogen Production: Costs, Policies, and Scalability in the Transition to Net-Zero. *Int. J. Hydrogen Energy* **2025**, *128*, 473–487. https://doi.org/10.1016/j.ijhydene.2025.04.013.

3. **EI Knowledge.** New Energy World Magazine. Energy Institute, 2024. Available online: https://knowledge.energyinst.org/new-energy-world/article?id=138706 (accessed 2025).

4. **Naphtha | 2005-2020 Data | 2021-2022 Forecast | Price | Quote | Chart | Historical.** Tradingeconomics.com, n.d. Available online: https://tradingeconomics.com/commodity/naphtha (accessed 2025).

5. **Malik, V.** *St James Parish Map, Louisiana*; MapsofWorld, April 3, 2025. https://www.mapsofworld.com/usa/states/louisiana/counties/counties-st-james-parish-map.html (accessed 2026-03-08).

REferences

6. **Occupational Safety and Health Administration**. *Carbon Monoxide*; OSHA Occupational Chemical Database; U.S. Department of Labor: Washington, DC. https://www.osha.gov/chemicaldata/462
7. **Occupational Safety and Health Administration.** *Carbon Dioxide*; OSHA Occupational Chemical Database; U.S. Department of Labor: Washington, DC. https://www.osha.gov/chemicaldata/183
8. **Occupational Safety and Health Administration.** *Oxygen*; OSHA Occupational Chemical Database; U.S. Department of Labor: Washington, DC. https://www.osha.gov/chemicaldata/920
9. **Occupational Safety and Health Administration.** *Kerosene (Jet Fuels)*; OSHA Occupational Chemical Database; U.S. Department of Labor: Washington, DC. https://www.osha.gov/chemicaldata/703
10. **Occupational Safety and Health Administration.** *VM&P Naphtha*; OSHA Occupational Chemical Database; U.S. Department of Labor: Washington, DC. https://www.osha.gov/chemicaldata/707

REfrences

11. U.S. Energy Information Administration (EIA). *Energy Costs.* https://www.eia.gov/electricity/annual/html/epa_02_10.html (accessed 2026).

12. U.S. Energy Information Administration (EIA). *Today in Energy: Daily Energy Prices.* https://www.eia.gov/todayinenergy/prices.php (accessed 2026).

13. Business Analytiq. *Oxygen Prices 2026 | February Price Chart, Trend & Forecast.* https://businessanalytiq.com/procurementanalytics/index/oxygen-price-index/ (accessed 2026).

14. Business Analytiq. *Green Hydrogen Price Index.* https://businessanalytiq.com/procurementanalytics/index/green-hydrogen-price-index/ (accessed 2026).

15. U.S. Congressional Research Service. *Sustainable Aviation Fuel Tax Credit (CFPC) Overview.* https://www.congress.gov/crs-product/IF12502 (accessed 2026).

16. ZipRecruiter. *Plant Operator Salary in Louisiana: Hourly Rate.* https://www.ziprecruiter.com/Salaries/Plant-Operator-Salary--in-Louisiana (accessed 2026).

References

17. **Dry, M. E.** *The Fischer–Tropsch Process: 1950–2000.* Catalysis Today, 2002.
18. **Speight, J. G.** *The Chemistry and Technology of Petroleum.* 5th ed. CRC Press, 2014.
19. **Ancheyta, J**. *Hydroprocessing of Heavy Oils and Residua.* CRC Press, 2007.
20. **Gary, J. H.; Handwerk, G. E.; Kaiser,** M. J. *Petroleum Refining: Technology and Economics.* 5th ed. CRC Press, 2007.
21. **Prudhomme, J**. *Upgrading – Hydrocracker/Hydrotreater – Run HTHC-PU-2011-2 – 3 Day Test Run.* Rentech, Inc., Technical Report, March 5, 2012.
22. **Towler, G.; Sinnott, R.** *Chemical Engineering Design: Principles, Practice and Economics of Plant and Process Design.* 3rd ed.; Butterworth-Heinemann: Oxford, UK, 2022.
23. **U.S. Department of Energy,** National Energy Technology Laboratory (NETL). *Gasifipedia: Syngas Contaminant Removal and Conditioning.* Overview. https://www.netl.doe.gov/research/coal/energy-systems/gasification/gasifipedia/cleanup
24. **Airlines for America.** *Sustainable Aviation Fuel Price Comparison.* Overview. https://www.airlines.org/dataset/sustainable-aviation-fuel-price-comparison/